

Intro to spatial 'omics

>
DNA
Keratin
CD20
CD3

Day 1

1. Histology
2. Spatial proteomics
3. Live demo

Day 2

1. Spatial transcriptomics
2. Live demo

But first: Spatial biology goes beyond biomedicine

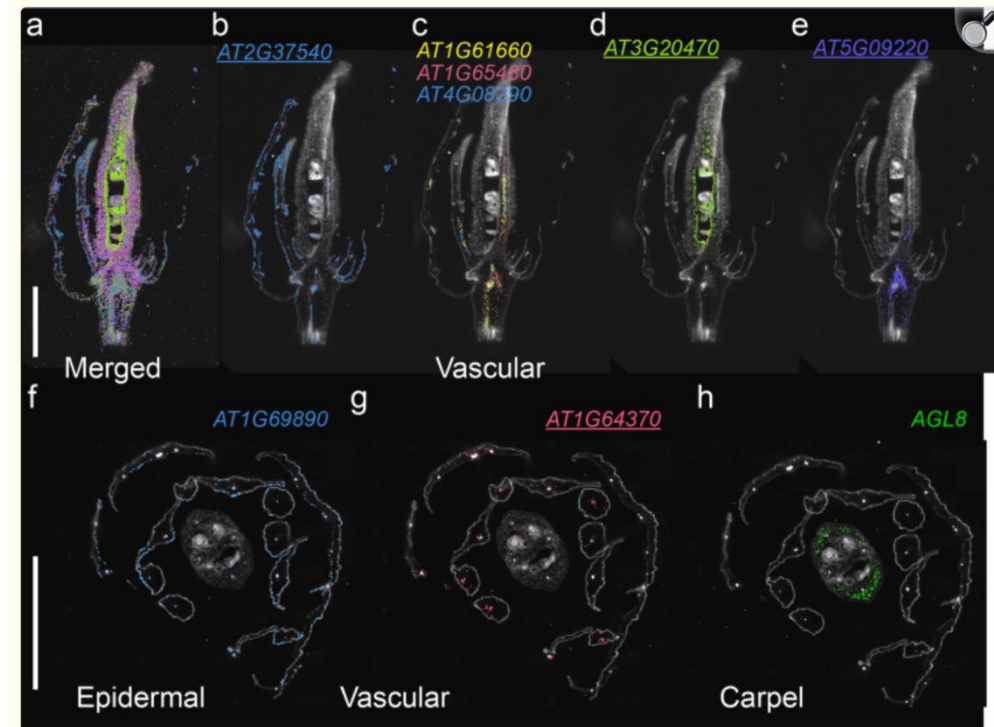
“First, the spatial context should be regarded as the central axis of interpretation, since identical changes in transcription levels yield divergent biological outcomes depending on their anatomical location, whether in the epidermis, endoderm, bundle sheath, or companion cells of the phloem.”

► Int J Mol Sci. 2025 Dec 7;26(24):11819. doi: [10.3390/ijms262411819](https://doi.org/10.3390/ijms262411819) 

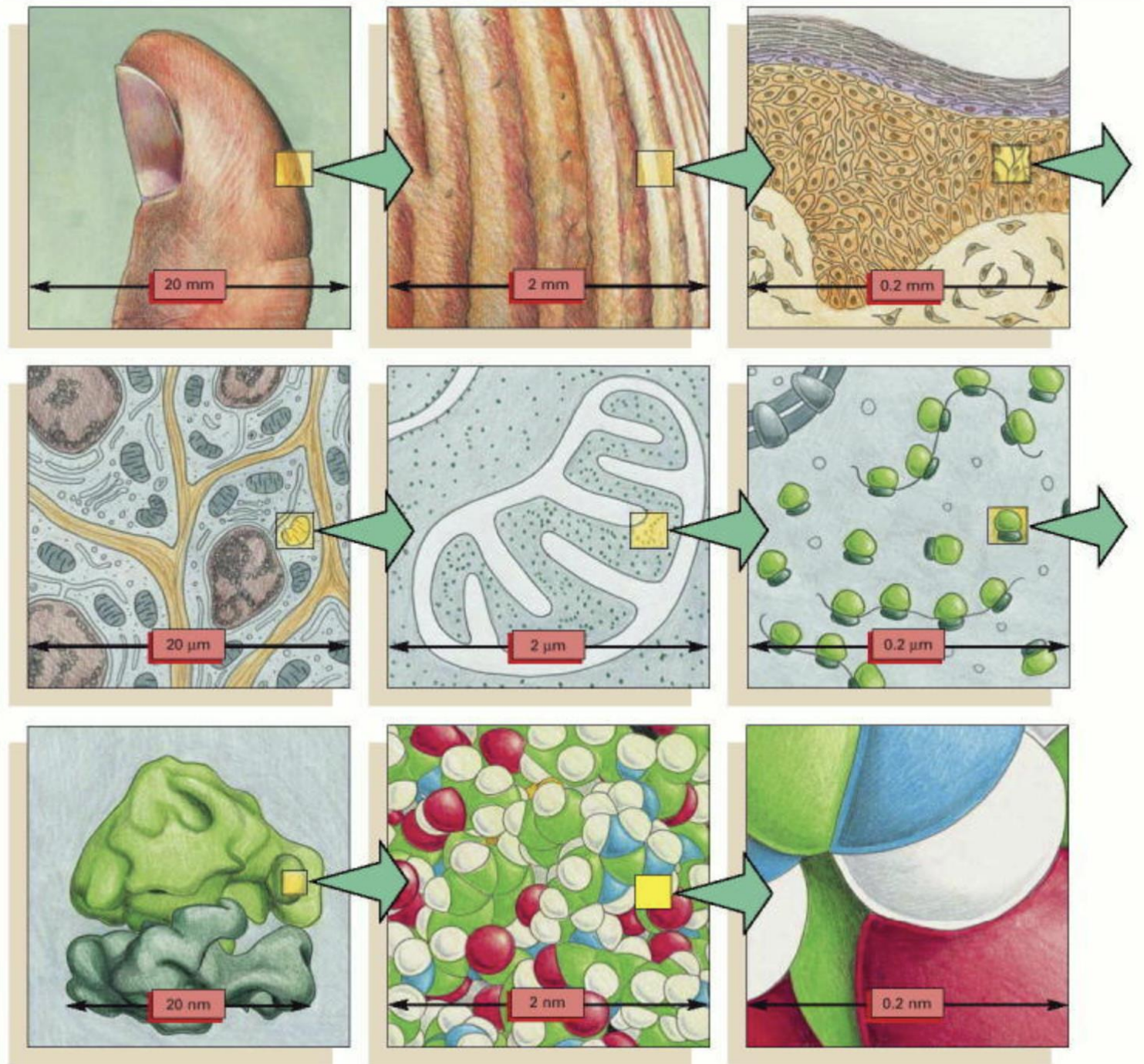
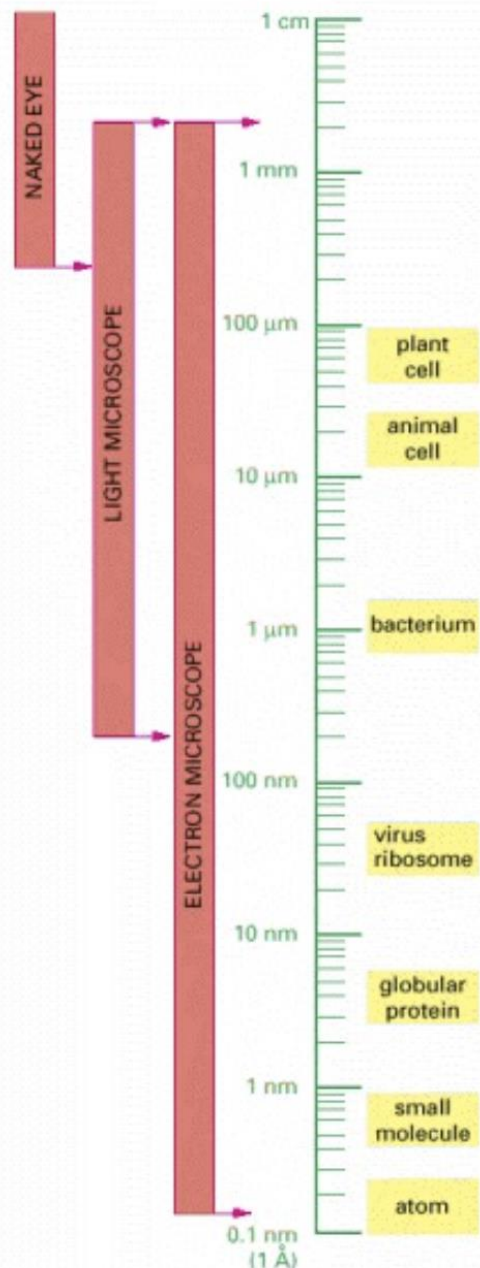
Why “Where” Matters as Much as “How Much”: Single-Cell and Spatial Transcriptomics in Plants

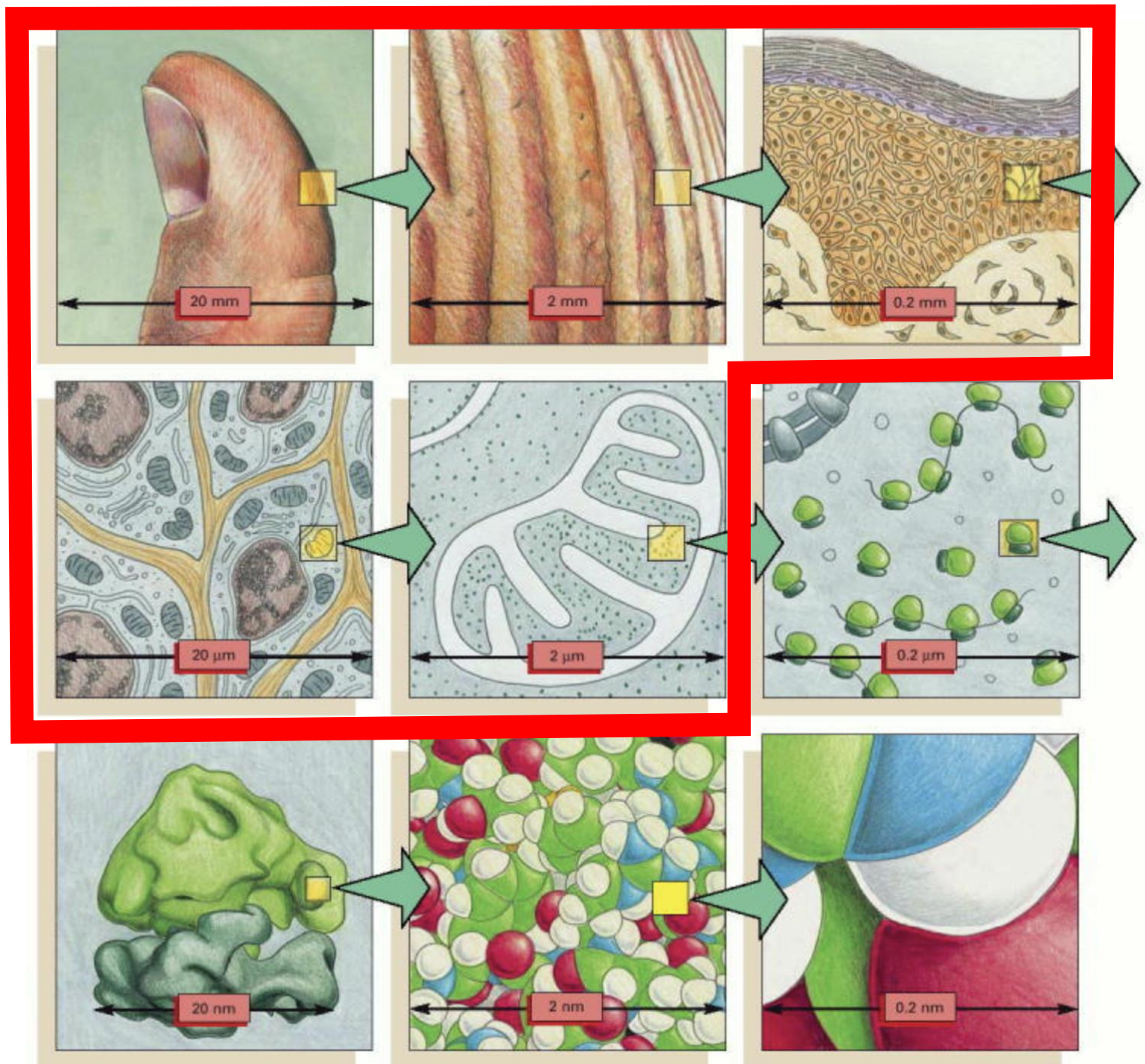
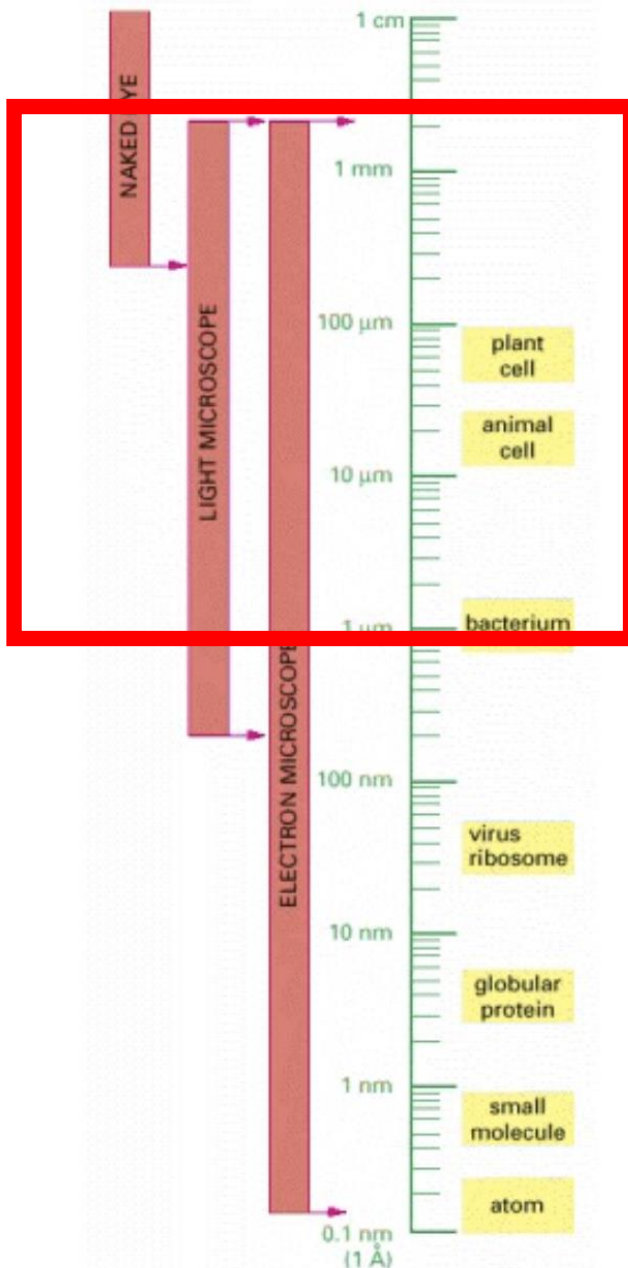
[Kinga Moskal](#)¹, [Marta Puchta-Jasińska](#)¹, [Paulina Bolc](#)¹, [Adrian Motor](#)¹, [Rafał Frankowski](#)¹, [Aleksandra Pietrusińska-Radzio](#)¹, [Anna Rucińska](#)^{1,2}, [Karolina Tomiczak](#)¹, [Maja Boczkowska](#)^{1,*}

Extended Data Fig. 5. Spatially resolved cell identities in the *Arabidopsis* flower.

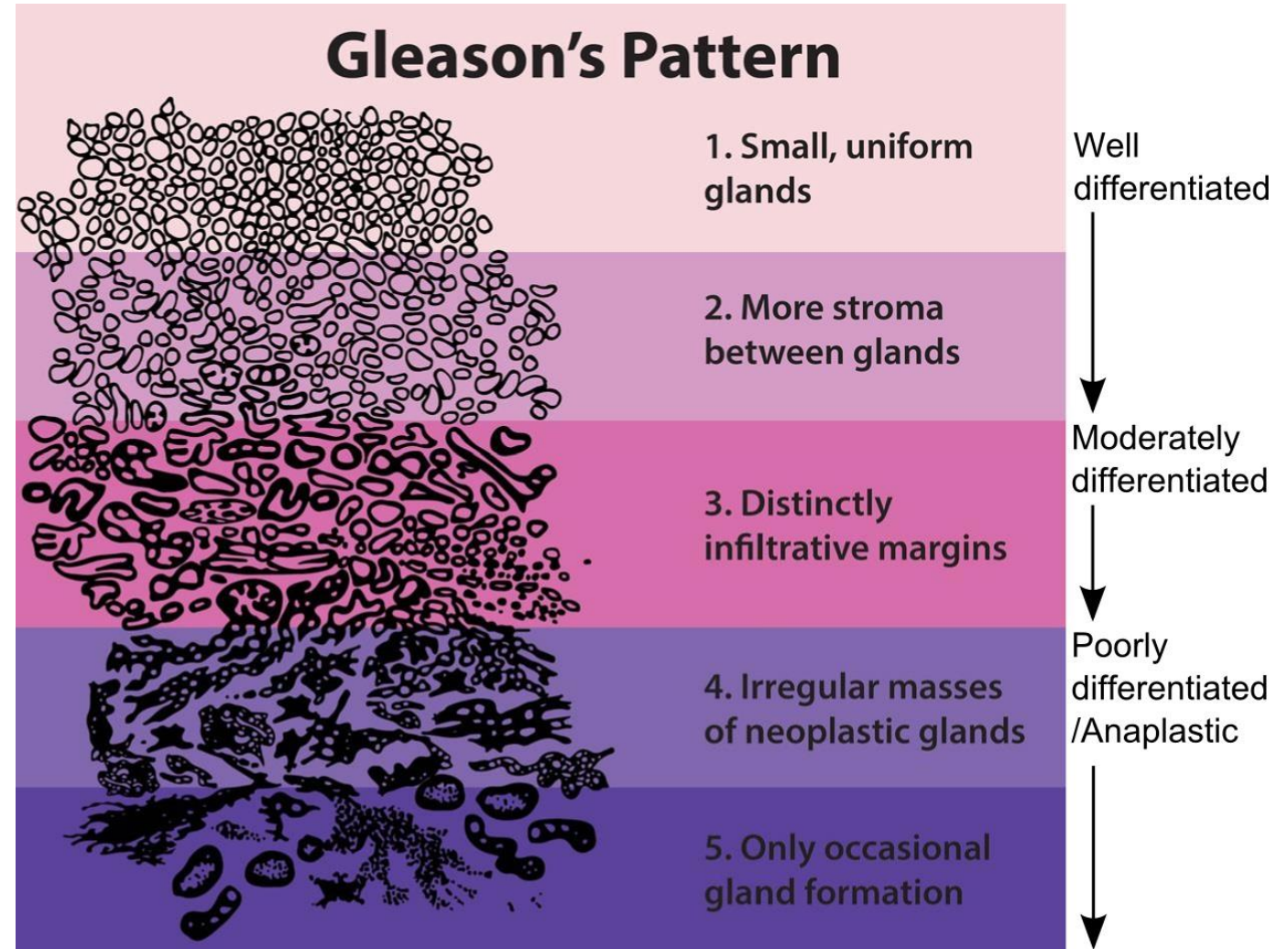
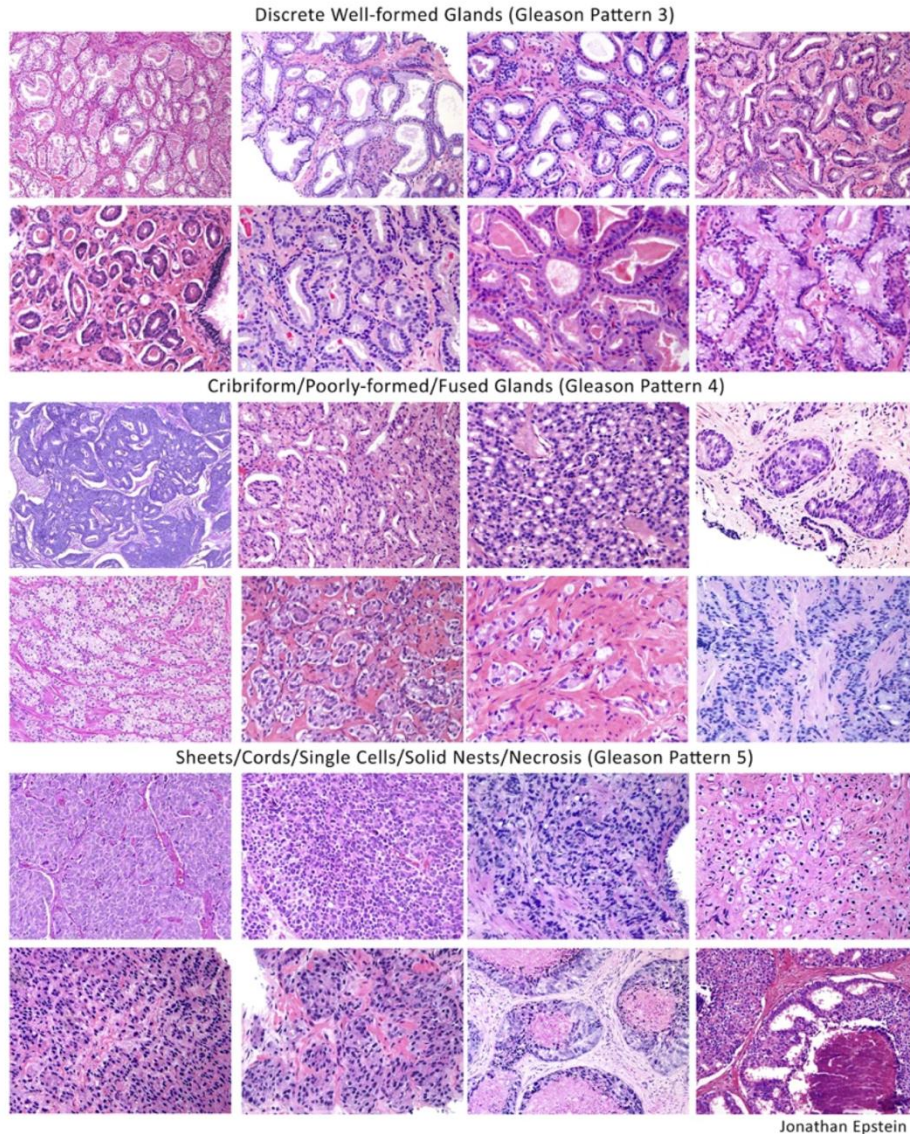


Lee et al. (2025) *Nature plants*

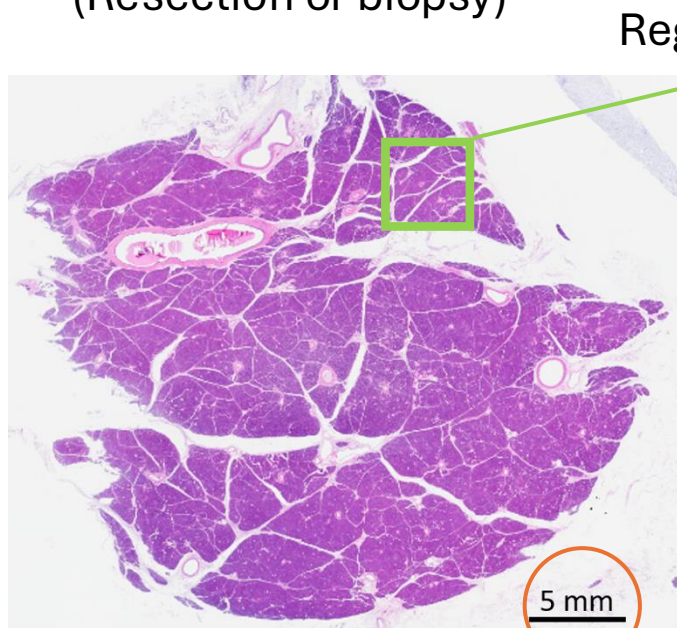
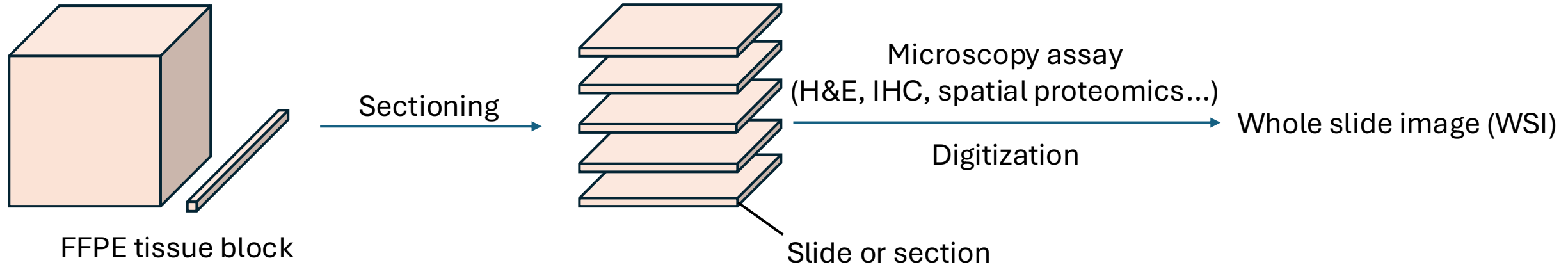




Hematoxylin & Eosin: The gold standard of histopathology

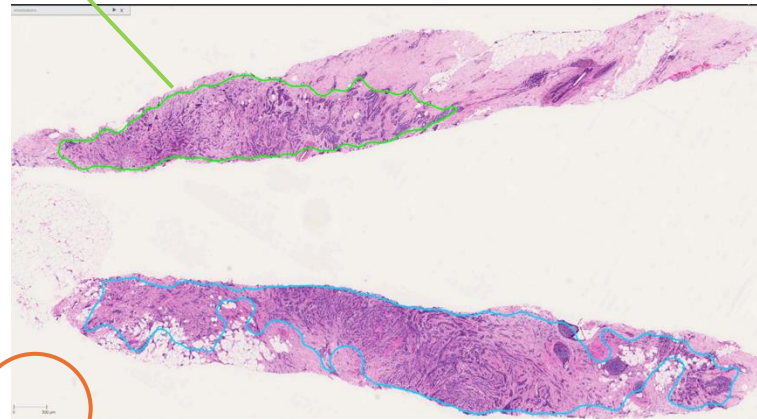


Tissue microscopy terms (as understood by a computational biologist)



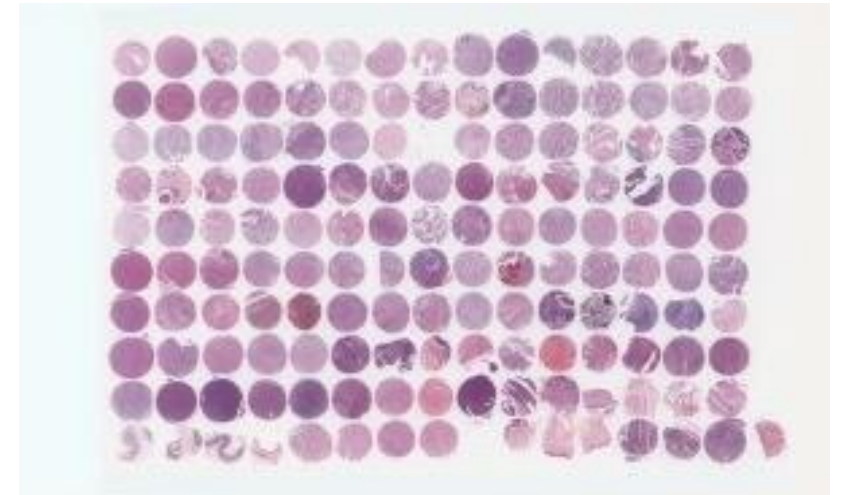
Regions of Interest (ROI)

Core needle biopsies



5 mm

Tissue microarray (TMA)

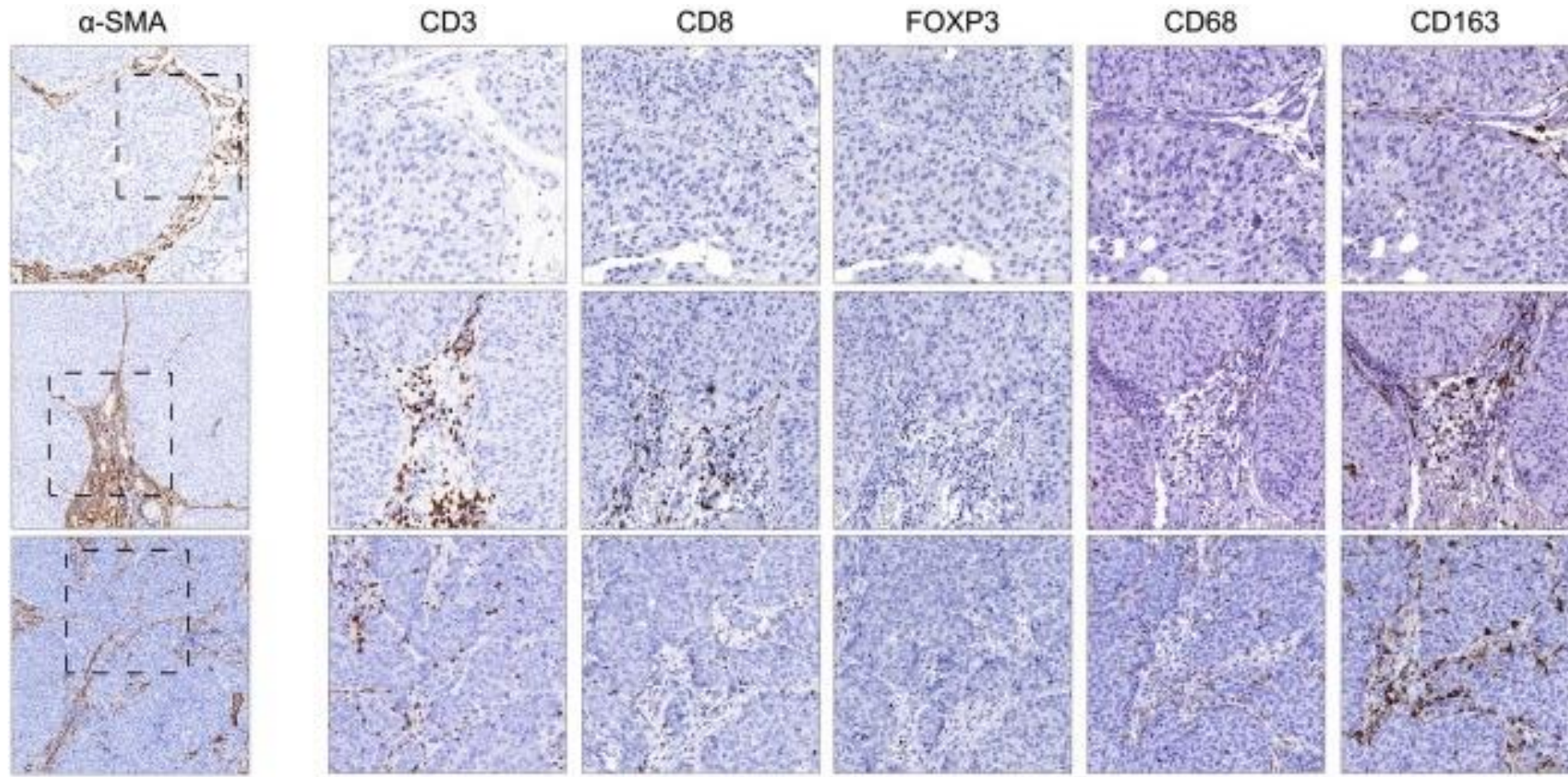


molecular.pathology.ufl.edu/category/molecular-pathology-core/microscope/digital-slide-scan/

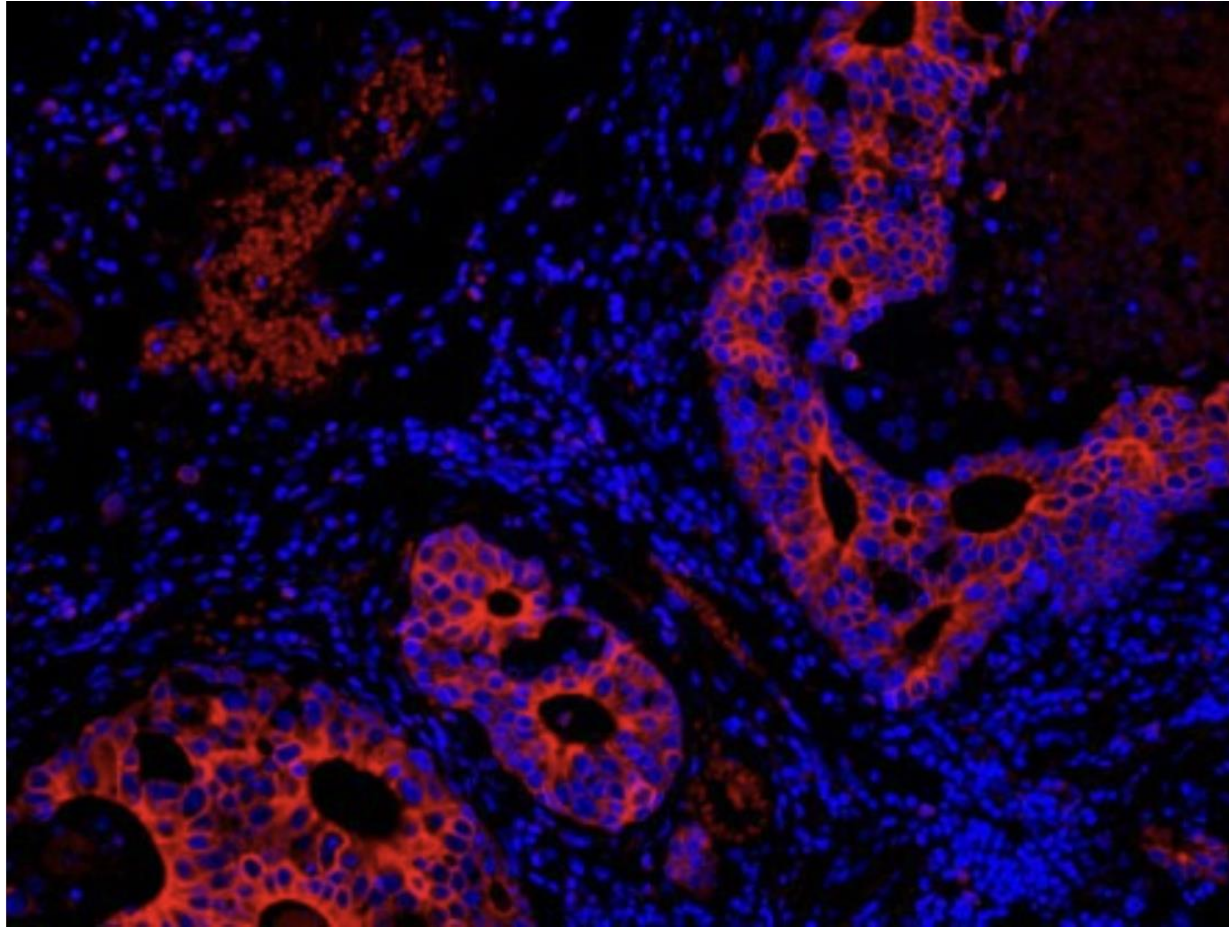
Focke, Decker, and Van Diest
(2016) *Histopathology*

brpc.duke.edu/brpc-special-services/tissue-microarray

Immunohistochemistry (IHC) staining complements H&E with molecular information



Fluorescent IHC allows for higher resolution and better quantification of protein abundance



“Immunofluorescence method for IHC detection” www.thermofisher.com

Spatial proteomics (multiplex tissue imaging)

nature methods

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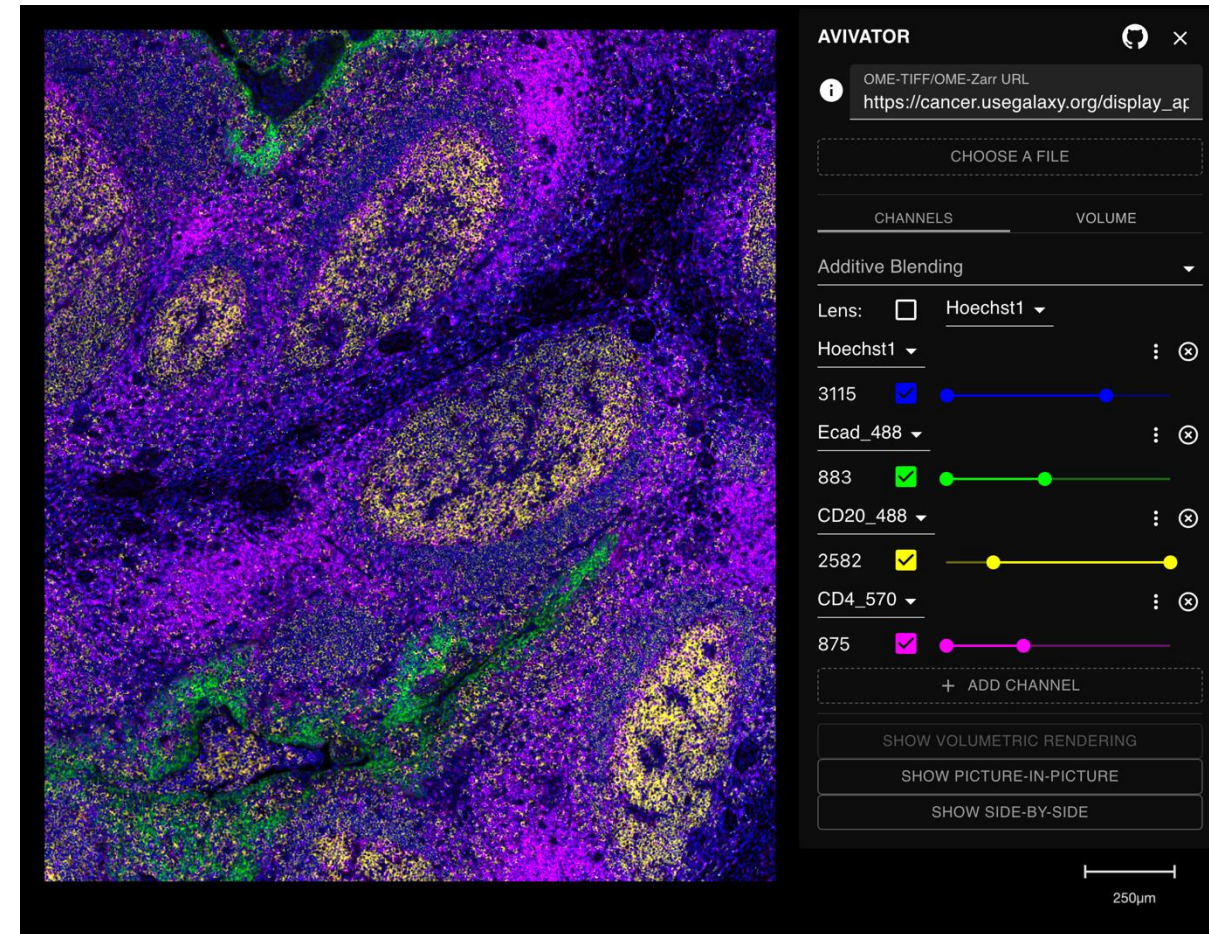
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Focus | 10 December 2024

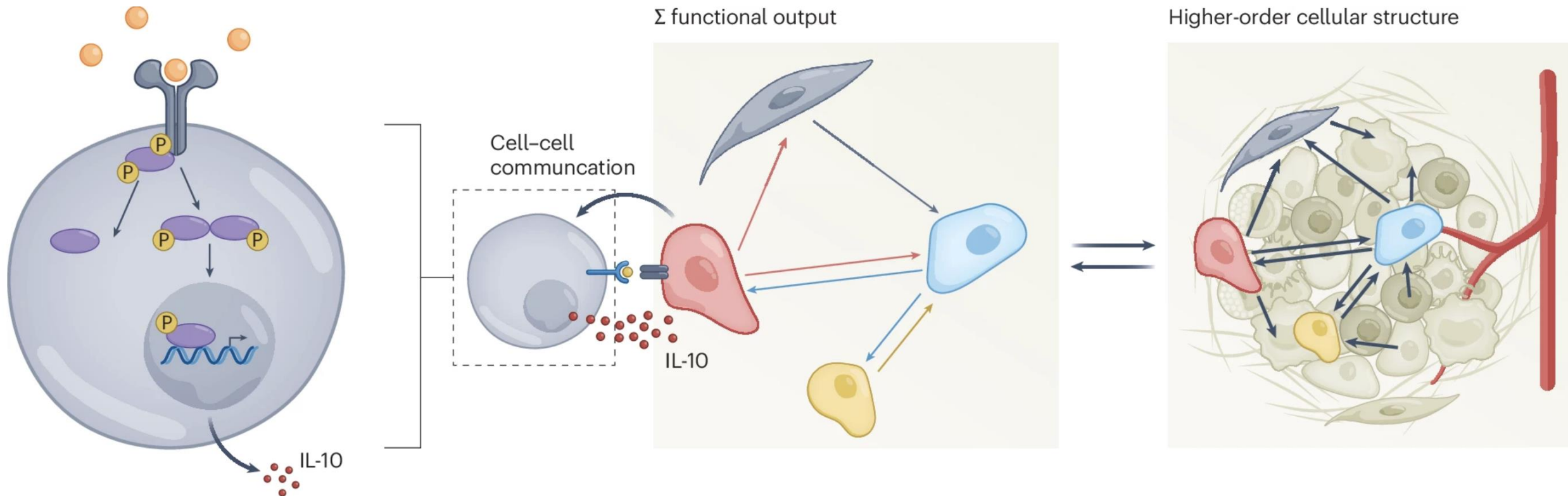
Method of the Year 2024: spatial proteomics

Spatial proteomics is our pick for Method of the Year 2024, for the impact that these technologies have had on the understanding of the organization, structure and function of complex tissues, including in global tissue atlas projects.



Why the method of the year in 2024?

From: [Highly multiplexed imaging in the omics era: understanding tissue structures in health and disease](#)



Why the method of the year in 2024?

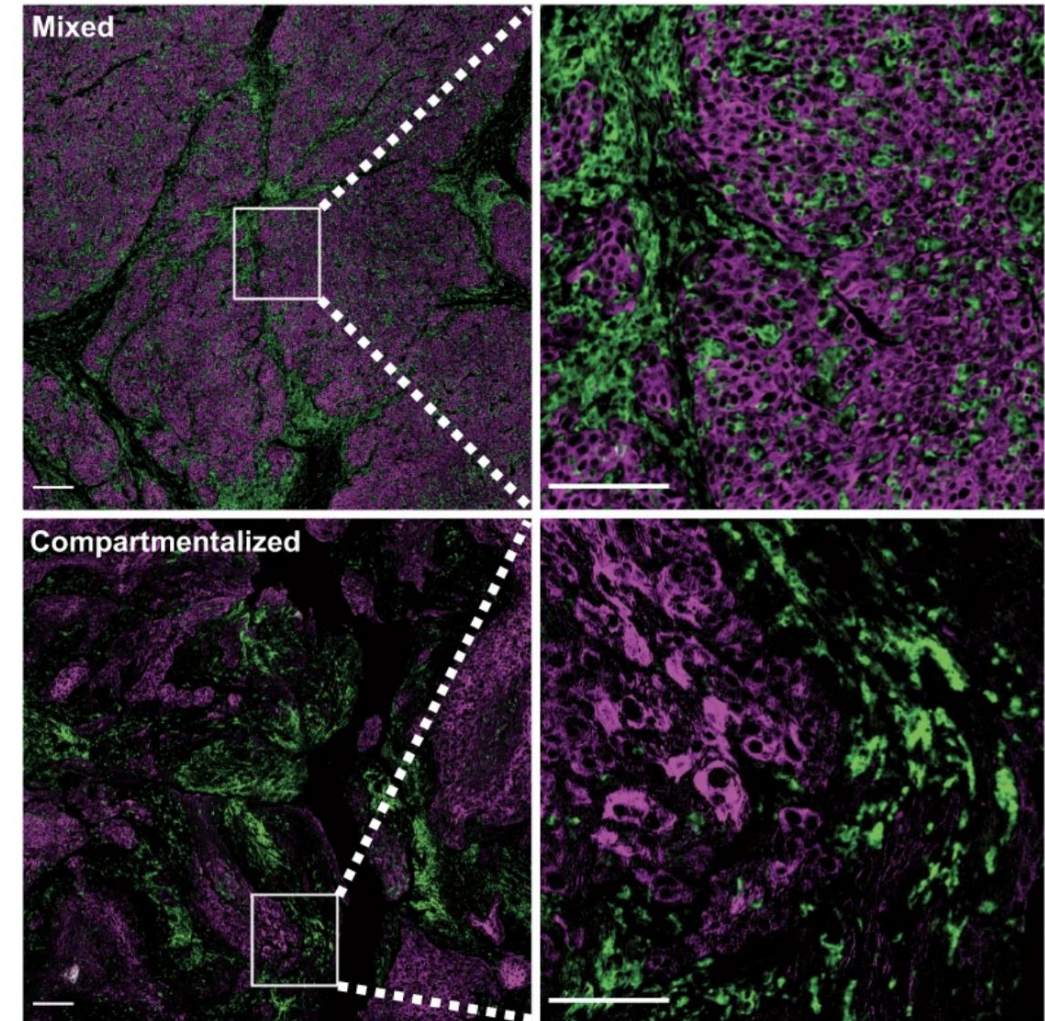
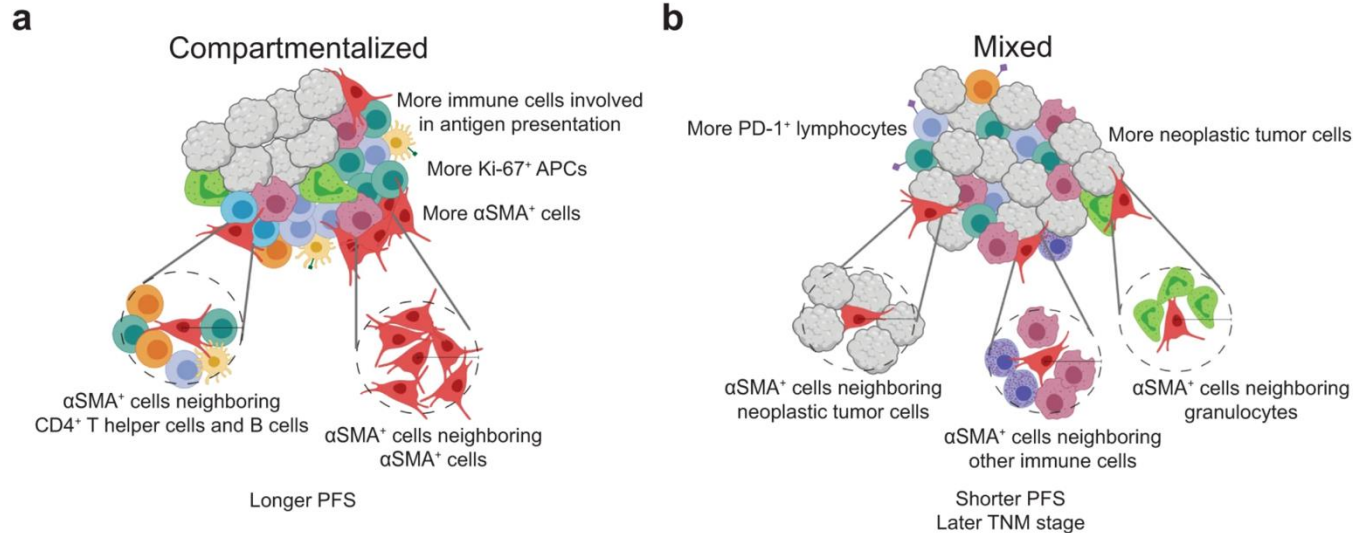
Article | [Open access](#) | Published: 25 February 2022

Single-cell spatial architectures associated with clinical outcome in head and neck squamous cell carcinoma

[Katie E. Blise](#), [Shamilene Sivagnanam](#), [Grace L. Banik](#), [Lisa M. Coussens](#) & [Jeremy Goecks](#) 

[npj Precision Oncology](#) **6**, Article number: 10 (2022) | [Cite this article](#)

8149 Accesses | **51** Citations | **9** Altmetric | [Metrics](#)

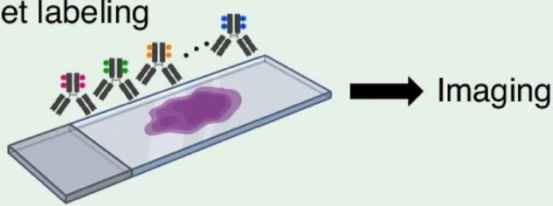


Spatial proteomics covers a lot of different technologies

From: [Spatial analysis by current multiplexed imaging technologies for the molecular characterisation of cancer tissues](#)

Single-shot imaging

Simultaneous target labeling



Mass spectrometry-based methods

Imaging Mass
Cytometry,
Hyperion

Multiplexed Ion
Beam Imaging,
MIBIscope

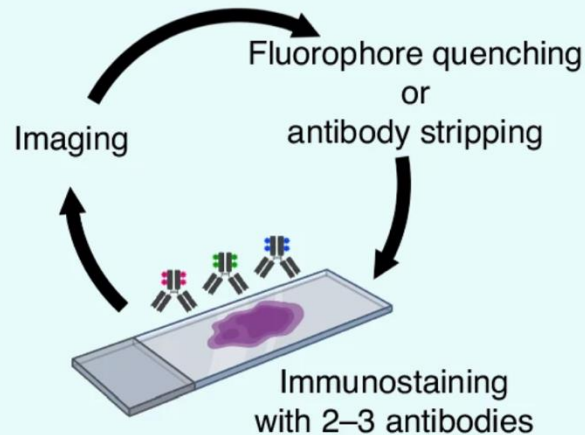
Optical microscope-based methods

Orion

Multicycle staining

Phenolmager

Multicycle imaging



DNA barcoded antibodies and reporter dyes

PhenoCycler
SignalStar

Direct IHC-based methods

MACSima

CellDIVE

CellScape

MELC

IBEX

MxIF

CyCIF

Indirect IHC-based methods

COMET

4i

SIFS

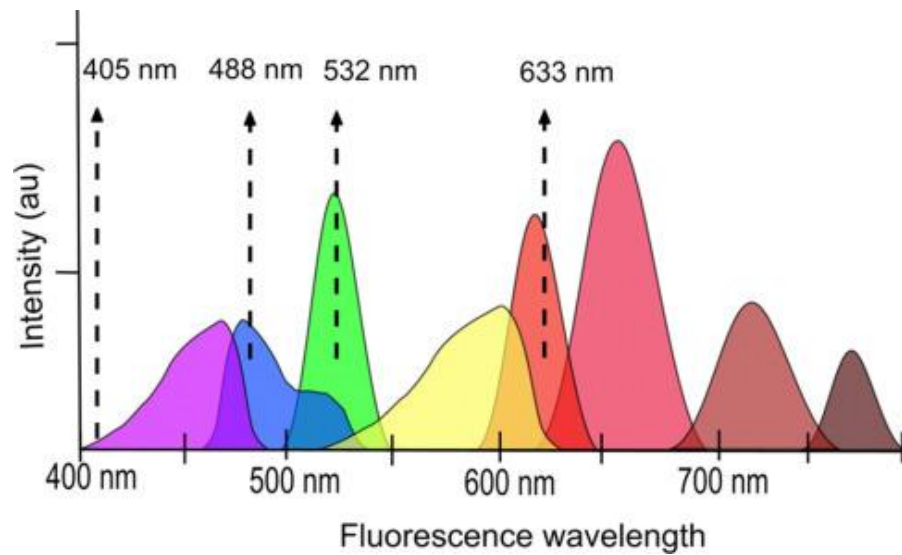
Chromogenic methods

mIHC

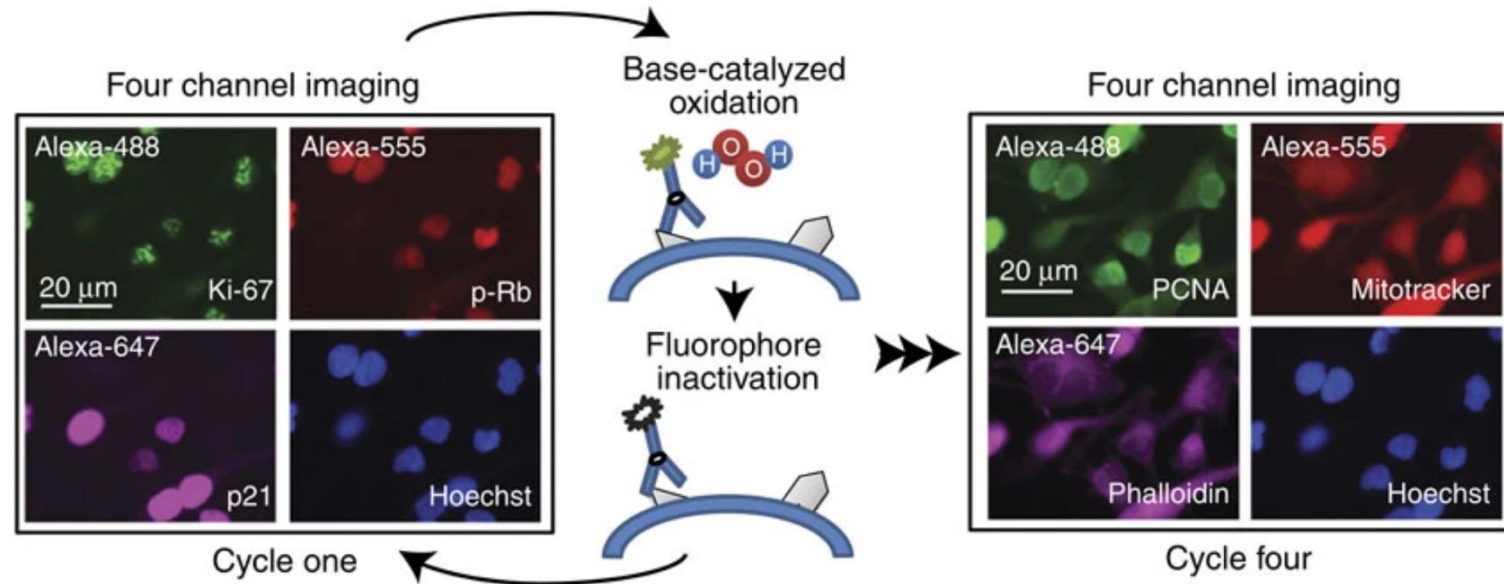
SIMPLE

MICSSS

Cyclic Immunofluorescence

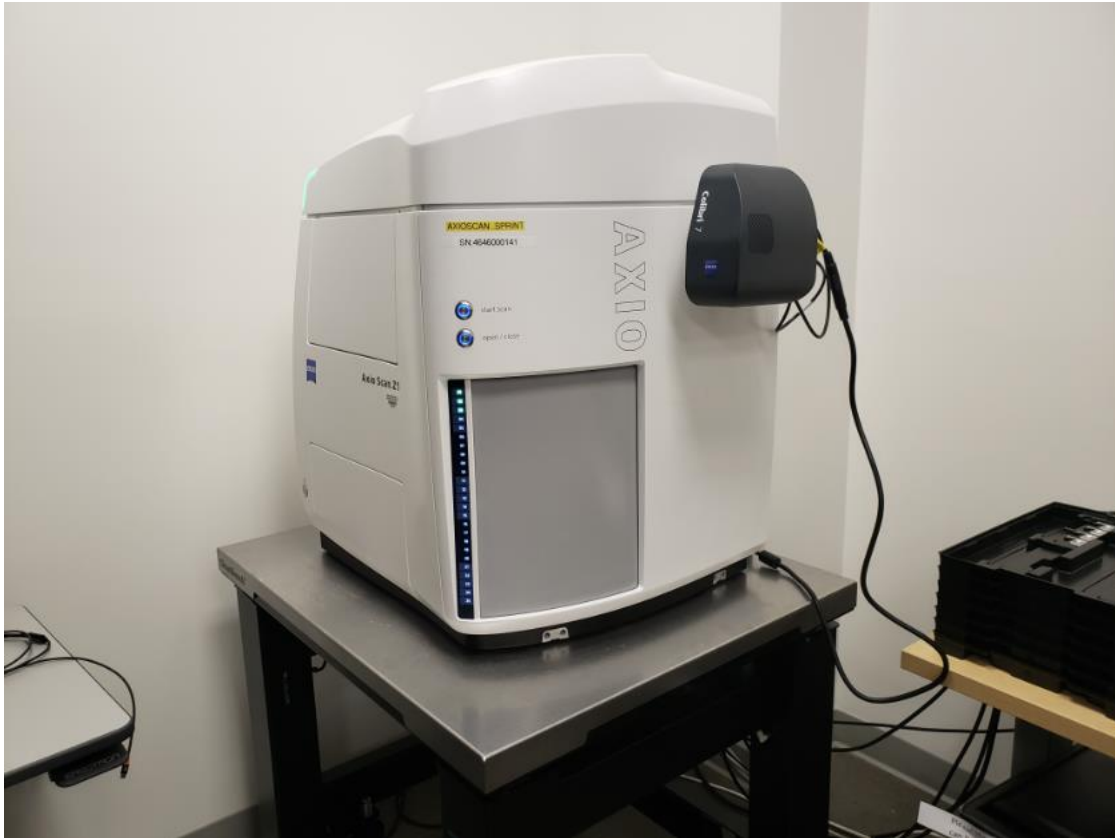


Bernas et al. 2006 *Mol. Cell. Proteomics*

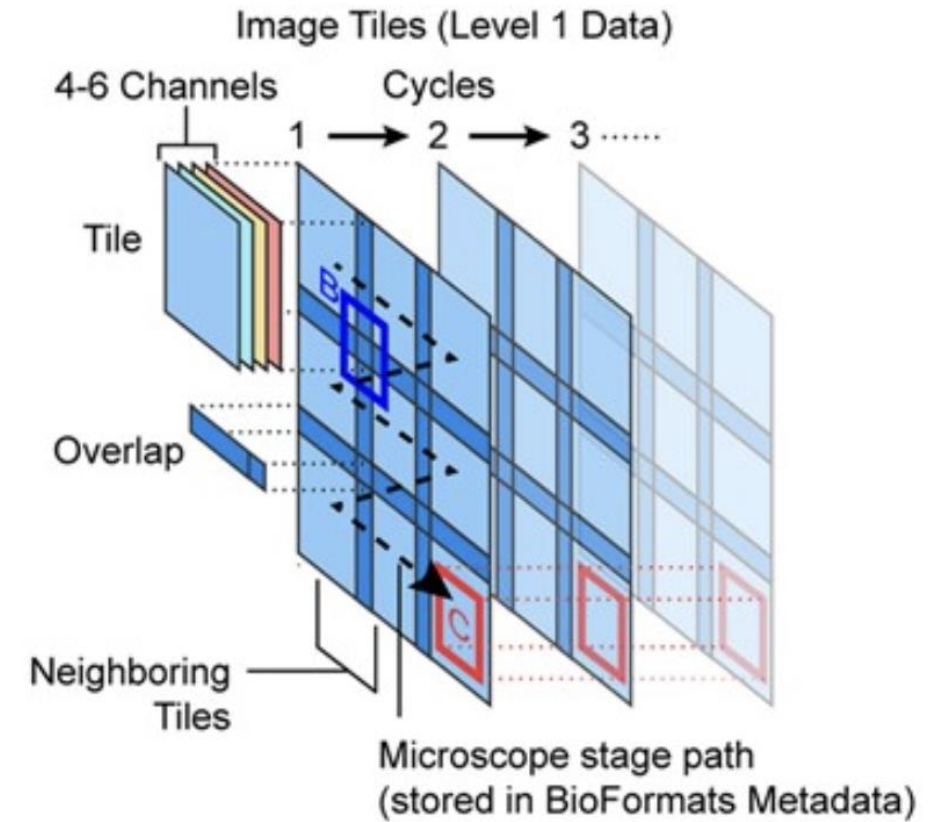


Lin et al 2015 *Nature Communications*

Slide scanners generate raw image tiles for each protein marker in the cycle

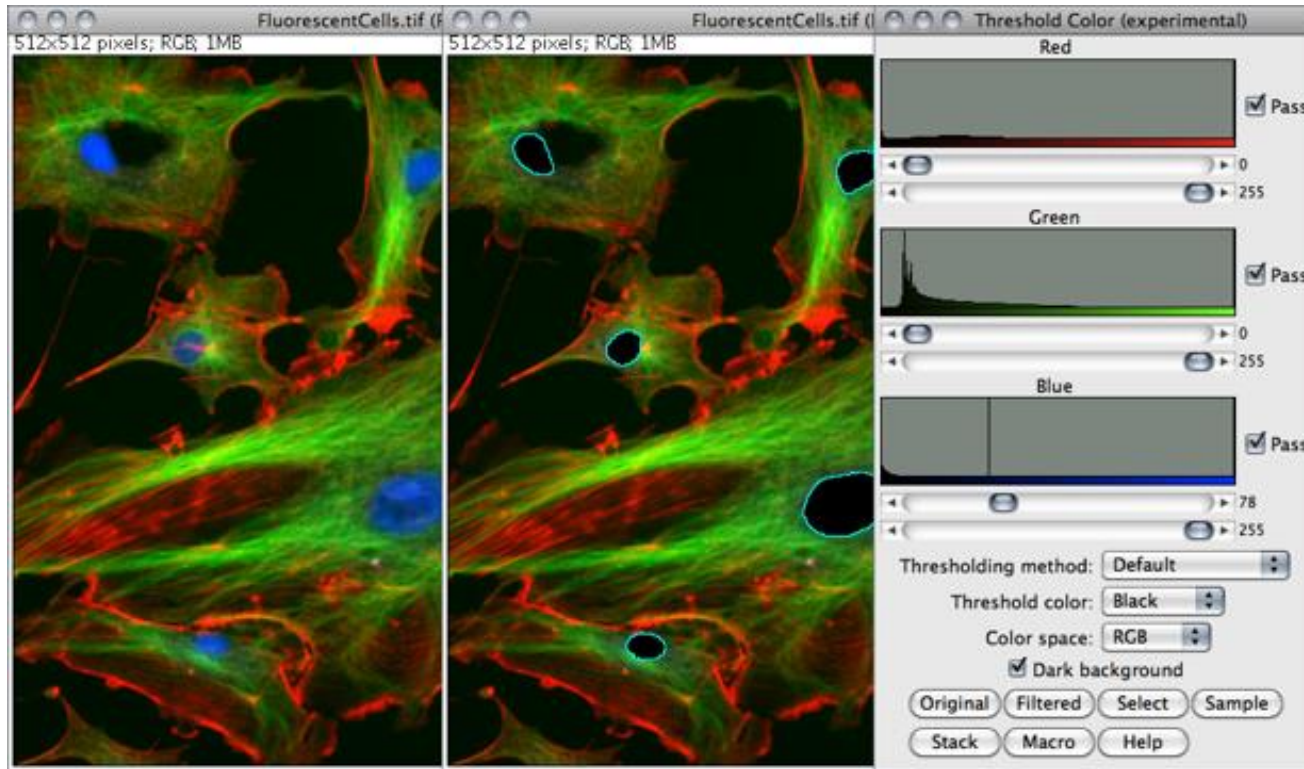


www.ohsu.edu/advanced-light-microscopy-core/zeiss-axioscan-7-slide-scanner



Muhlich et al. 2022 *Bioinformatics*

Manual processing has been the status quo,
but can introduce bias and does not scale

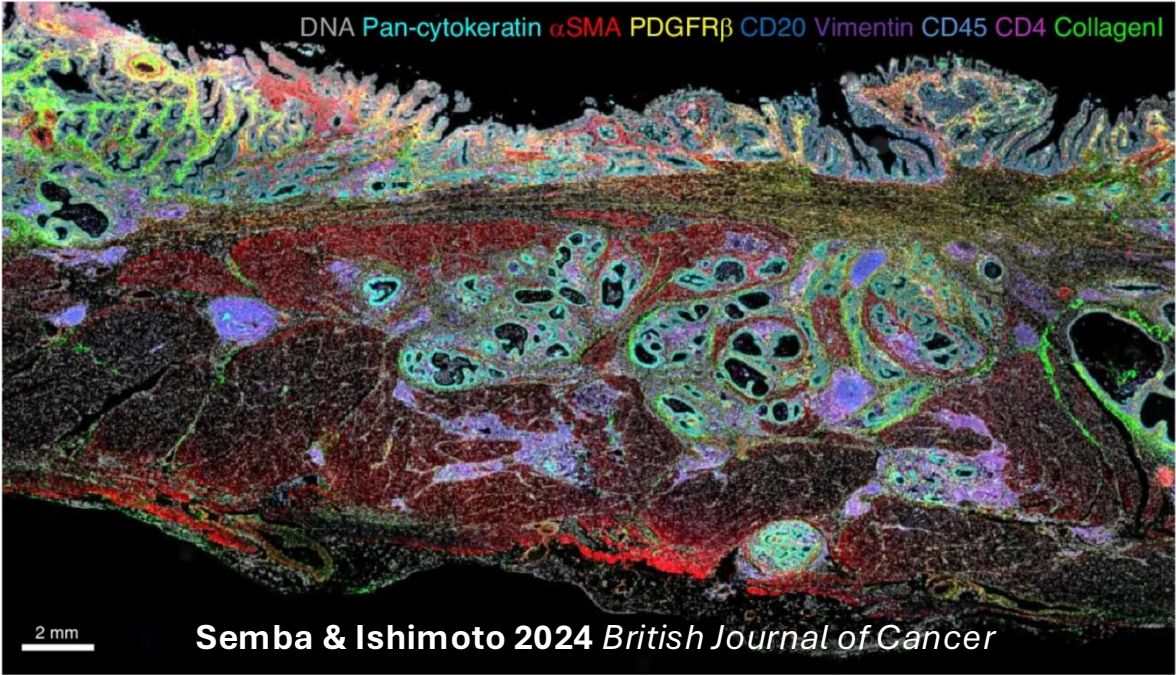


ImageJ

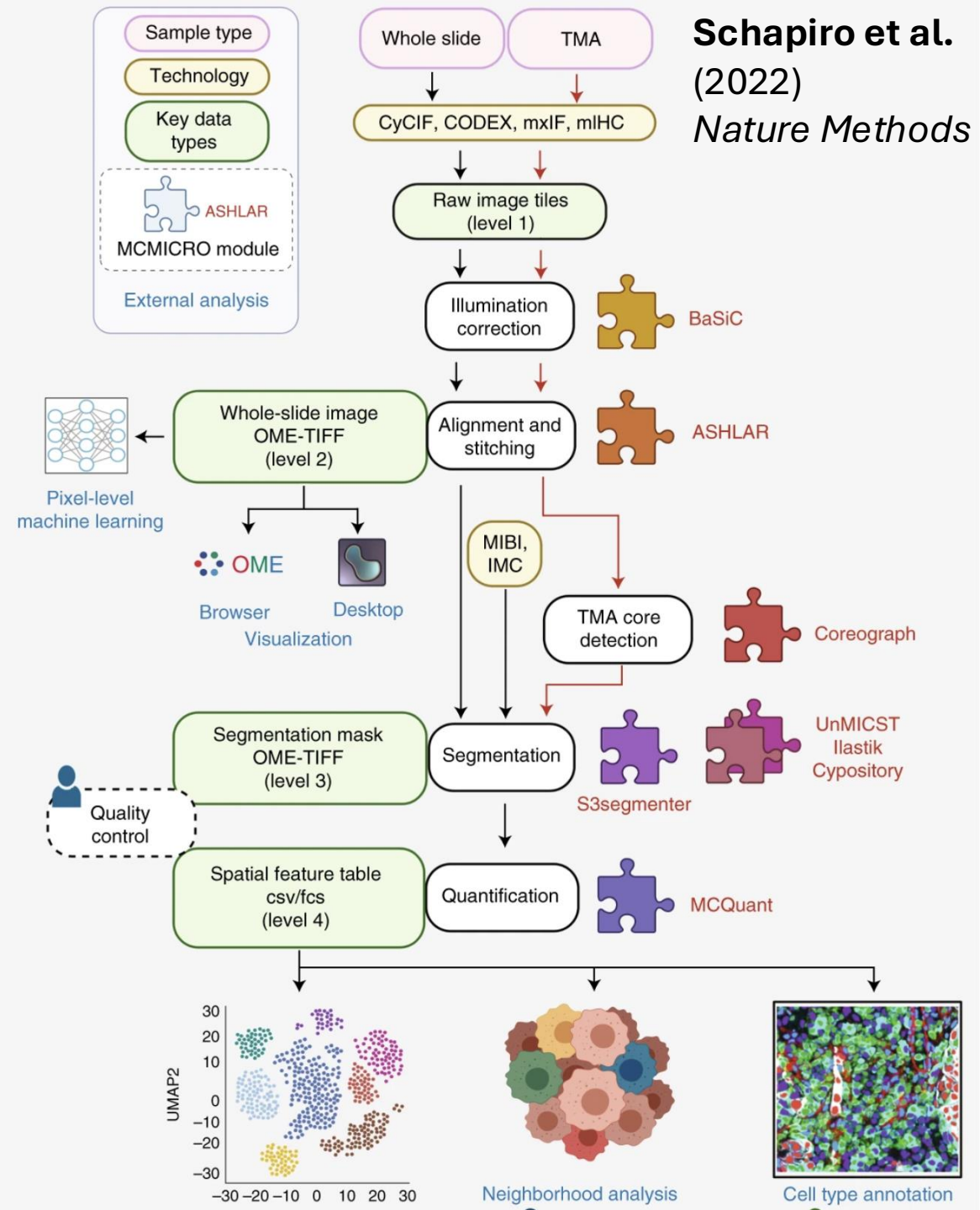


Photoshop

Overview of computational processing: images to cell feature table

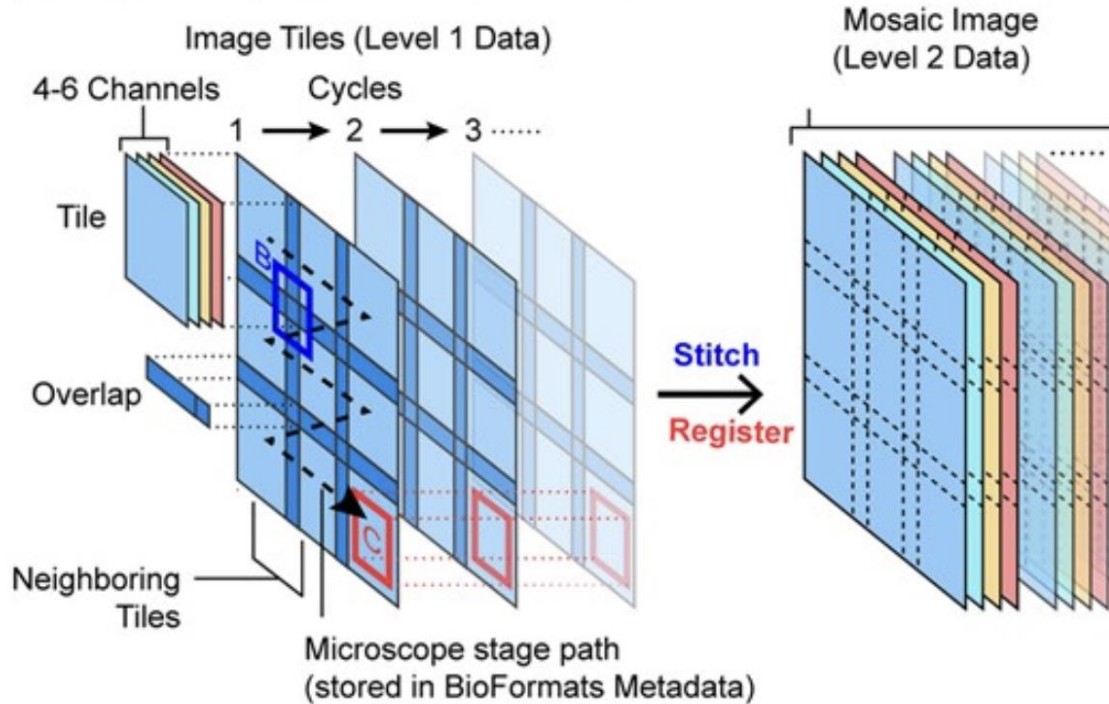


CellID	DNA	PanCK	aSMA	PDGFR	CD20	...	X_centroid	Y_centroid
0						...		
1						...		
2						...		
...						...		

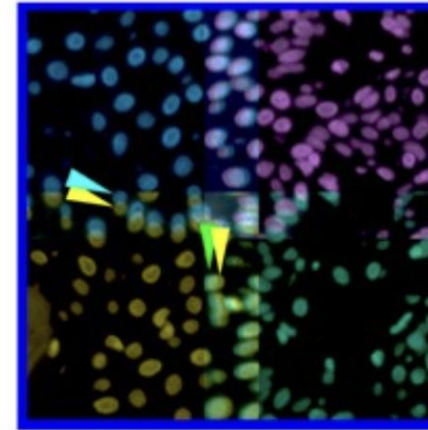


Microscope slide scanner produces one tiled image per round that need to be stitched & registered

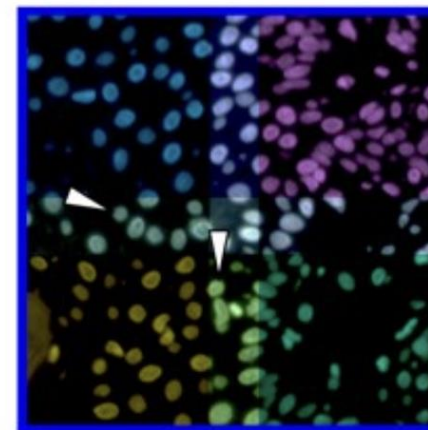
(a) Steps in imaging and data processing



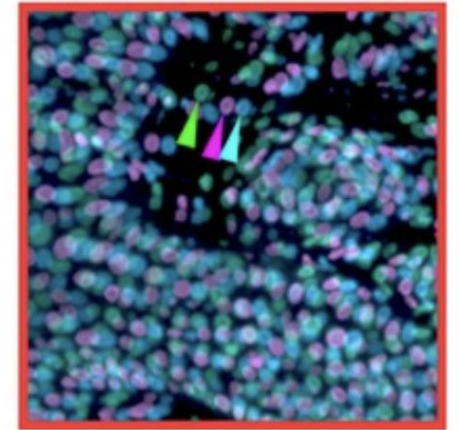
(b) Stitching four tiles



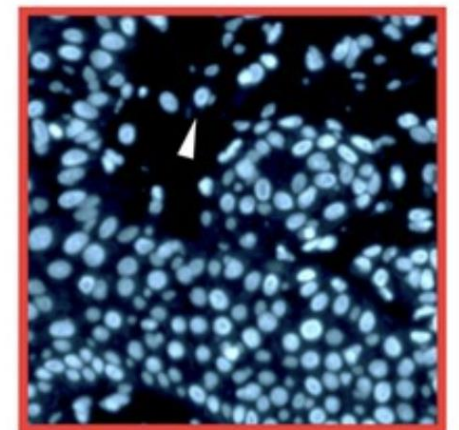
↓ Stitching



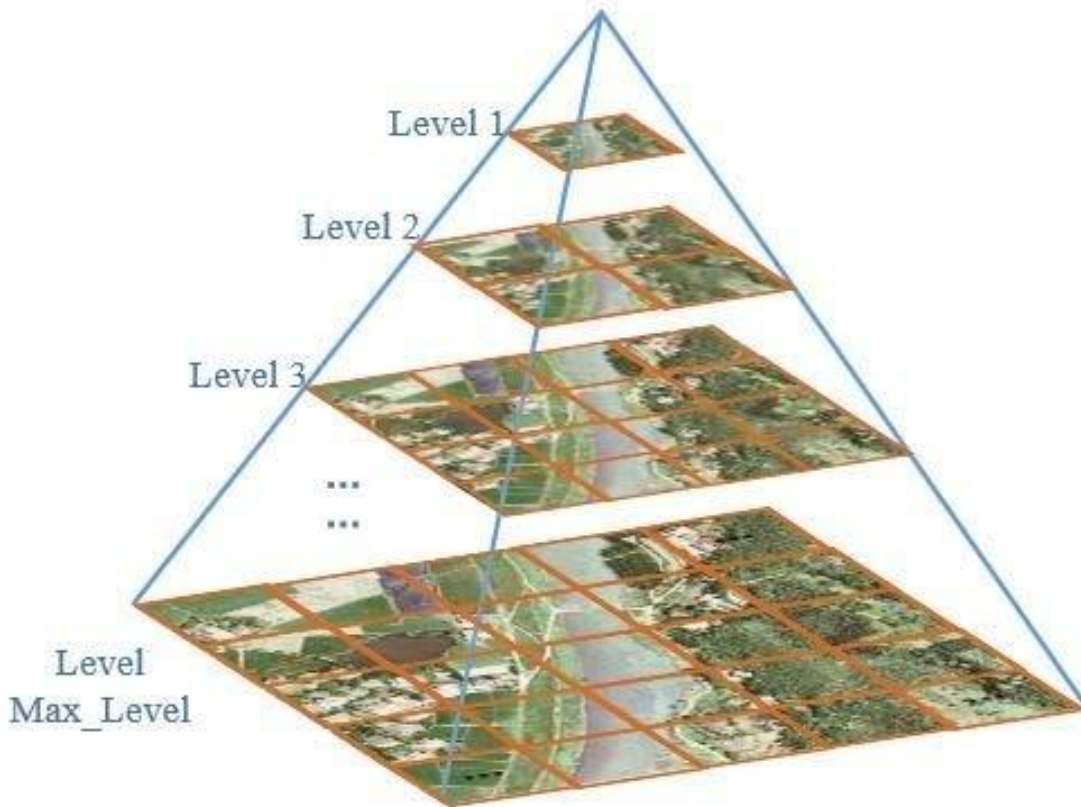
(c) Registering three cycles



↓ Registration



Pyramidal TIFF structure enables interactive visualization of massive images



Guo et al. (2016) *Advances in Electrical and Computer Engineering*

OME-TIFF

- An open data format for biomedical images
- Specifies structure and metadata
- Widely supported across biomedical imaging software

docs.openmicroscopy.org/ome-model/5.6.3/ome-tiff/

Next generation file formats

Brief Communication | [Open access](#) | Published: 29 November 2021

OME-NGFF: a next-generation file format for expanding bioimaging data-access strategies

Histochemistry and Cell Biology (2023) 160:223–251
<https://doi.org/10.1007/s00418-023-02209-1>

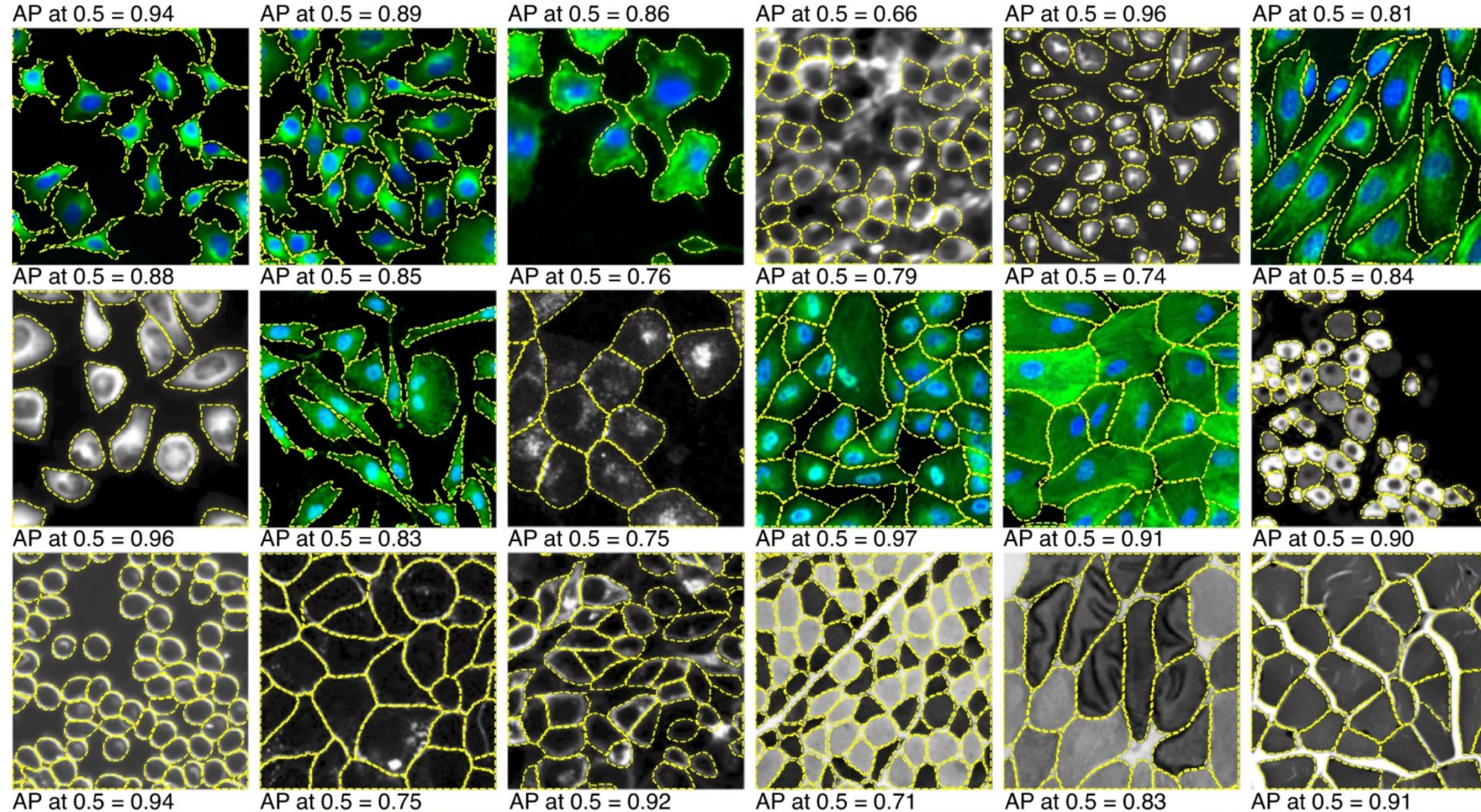
ORIGINAL PAPER



OME-Zarr: a cloud-optimized bioimaging file format with international community support

Nuclear & cellular segmentation defines cell boundaries

From: [Cellpose: a generalist algorithm for cellular segmentation](#)



Stringer et al. 2020 *Nature Methods*

The feature table is the starting point for most analyses

CellID	Hoechst1	anti_NaKATP	anti_CD3	anti_CD45RO	Ki67_488	Keratin_570
1	13673.167	3979.4167	914.08333	3444.583333	2842.3333	389.66667
2	13990.579	4874.2105	772.05263	2675.736842	2606.7895	1272.9474
3	10669.538	12600.231	938.46154	1149.846154	4988.9231	2073.3846
4	12799.917	3279.75	4803.5833	2860.75	2320.0833	311.08333
5	16906.704	5358.9259	16498	9213.333333	2852.3333	328.25926
6	18506.385	3474.359	6746.359	3828.102564	2702.2051	305.69231
7	8441	4879.9333	2175.2	1753.333333	2992.2	274.2
8	13036.8	3349.1	1079.9	1629.1	2490.8	355.85
9	13605.042	2826	1138.125	2688.041667	2388.9583	357.91667
10	17670.364	3020.0909	912.68182	1321	2577.5909	500.59091
11	13516.636	3582.4545	935.81818	1285.454545	2811.8182	368.45455
12	11877.644	3311.6889	997.68889	2274.111111	3334.6889	397.15556
13	15454.679	4524.7143	844.82143	1443.5	2600.3929	393.10714
14	17290.63	4545.6296	1002.463	1566.425926	2722.8148	711.03704
15	13821.794	12351.286	923.03175	2975.888889	5455.9524	3350.873
16	13956.615	6226.7846	924.49231	2770.784615	3725.5231	4616.1692
17	13809.2	5538.6111	819.16667	1674.577778	3057.5111	7940.3222
18	10092.017	11370.831	811.38983	1131.79661	5173.4746	2886.8983
19	10114.138	5042.5172	1034.8621	1046.034483	2878.8966	863.62069
20	6150.5333	5288.6	1954.2	1717.866667	3724.6667	23166.4

X_centroid	Y_centroid	Area	MajorAxisLength
245	1.5	12	4.472135955
590.89474	1.4210526	19	5.544698234
642.15385	1.4615385	13	4.36166456
2052	1.5	12	4.472135955
2070.8889	1.5555556	27	7.811725554
2116.7436	2.025641	39	8.593214978
2745.4	1.6	15	4.618802154
104.65	2.7	20	6.211554639
123.5	1.8333333	24	6.218252702
201.31818	2.0909091	22	5.774849788
230.36364	2.5454545	11	5.189483705
297.35556	2.6444444	45	8.679046651
334.07143	2.2857143	28	6.629945254
365.07407	3.0185185	54	9.466923189
495.09524	2.8571429	63	10.90503356
606.10769	2.9076923	65	11.07245689
843.57778	2.8777778	90	18.0454577
941.77966	2.8983051	59	10.50114087
1117.7586	2.4137931	29	6.697572962

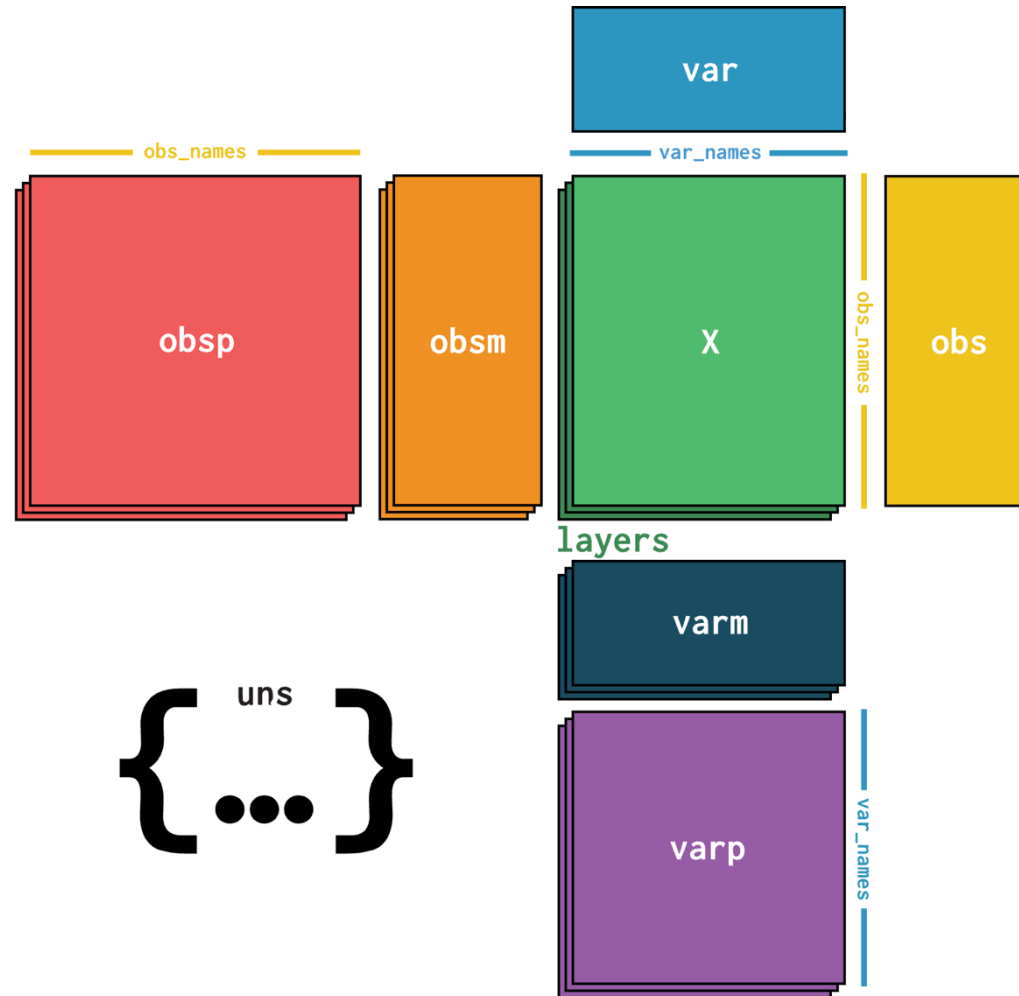
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...

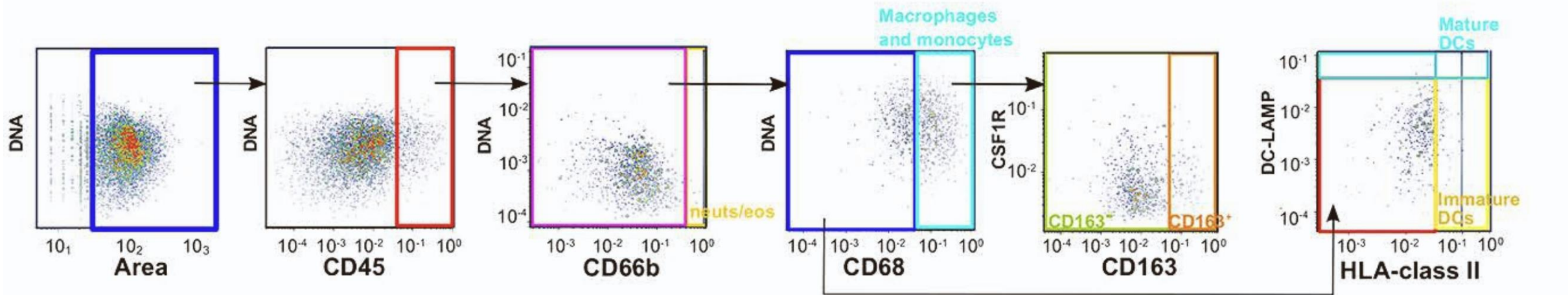
⋮

anndata:

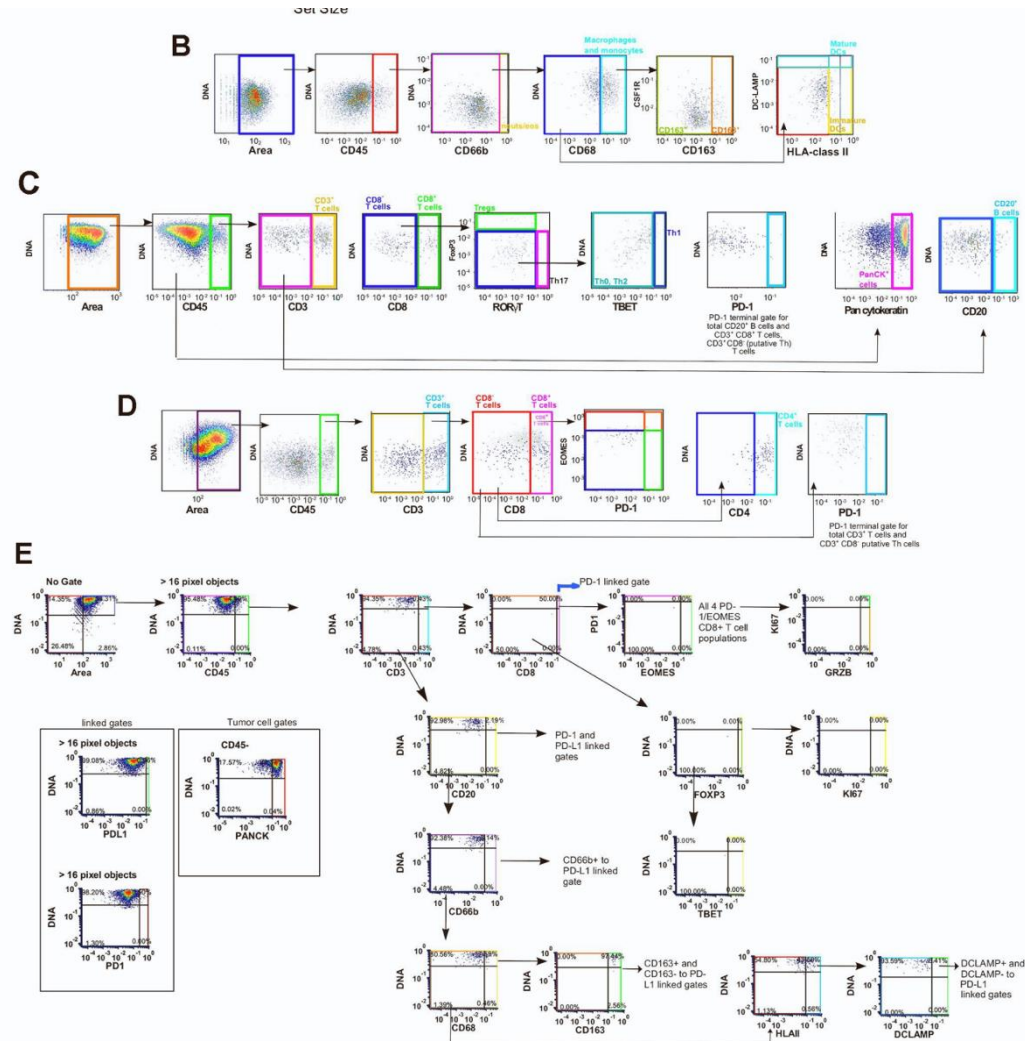
A better way to store the feature table and lots of other stuff



Cell type assignment: Gating



Cell type assignment: Gating



Cell type assignment: Unsupervised clustering

